

Solana Hybrid Sniper Ultra - Performance Analysis Report

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1. Executive Summary

This report analyzes the performance of the Solana Hybrid Sniper Ultra trading system based on 1,000 completed paper trades (session logged in /logs/app.log). The system currently runs in paper-trading mode with a micro-wallet (0.01 SOL starting balance). We project expected performance for 0.5-1.0 SOL starting capital with exponential compounding, incorporating realistic live-market degradation factors.

Key Findings at a Glance

Paper Win Rate:	81.2% (812 wins / 188 losses)
Paper Total PnL:	+0.9008 SOL on 1,000 trades
Avg Win / Loss:	+19.51% / -14.24% per position
Position EV:	+13.16% per trade on position size
Current Settings:	0.015 SOL max position, 1 concurrent position
Projected Live EV (1 SOL):	~+0.085% per trade on wallet (+1.7%/day)
Median 1-Year Outcome (1 SOL):	~2.1 SOL (+110%) with 25% chance of loss

2. Paper Trading Performance (Actual 1,000 Trades)

2.1 Aggregate Statistics

Metric	Value
Total Trades	1,000 (all closed)
Winning Trades	812
Losing Trades	188
Win Rate	81.2%
Total PnL (SOL)	+0.9008 SOL
Starting Balance	0.01 SOL
Final Balance	~0.017 SOL
Avg Win (position)	+19.51%
Avg Loss (position)	-14.24%
Best Trade	ROAF: +497.94% (SNIPER)
Worst Trade	Uhbs: -99.00% (SNIPER)
Session Duration	~1.7 hours (6,080s)
SOL/Hour (paper)	~0.0041 SOL/hr

2.2 Exit Reason Distribution (from logs)

Exit Reason	Description	Frequency
PNL_STAGNANT	No price movement for 60s at peak	High
PROFIT_DECLINE	Peak profit dropped >4%	High
EARLY_CUT	Loss exceeded half-stop threshold	Medium
LIQUIDITY_EXHAUSTION	5m volume / liquidity > 7x	Medium
LIQ_COLLAPSE	Liquidity dropped >50% or < \$300	Medium
TRAIL_STOP	Trailing stop at +15% peak, -8% trail	Several

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MAX_HOLD	Time limit reached (420-2100s)	Few
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2.3 Strategy Mode Breakdown

The session operated almost entirely in SNIPER mode due to micro-wallet constraints (balance < 0.05 SOL triggers micro-wallet mode with min position 0.003 SOL, max 0.006 SOL). Current settings for 0.5-1 SOL would enable all three modes with larger position sizes.

Mode	Trigger Conditions	Position Size	Typical Hold	Session %
SNIPER	Age<300s, Liq>\$10k, BP5m>65%	0.003-0.006 SOL	~800-1400s	~95%
MG	Vol mom>1.35, Price5m>1.2%, TX>8	0.003-0.008 SOL	Shorter	~3%
HWR	BP5m>70%, BP1h>51%, Liq>\$5k	0.003-0.008 SOL	Longer	~2%

3. Shadow Mode Analysis (Real Jupiter Quotes vs Paper)

The system runs shadow trades in parallel - real Jupiter quotes for every paper trade. This reveals the true execution cost gap. Current session results:

Metric	Paper	Shadow	Gap
Avg PnL/Trade	+24.18%	+25.24%	-1.06% (paper conservative)
Avg Price Impact	Model estimated	0.16%	Very low - Pump.fun direct
Avg Quote Fetch	N/A	454ms	Good
Avg Route Hops	N/A	1.0	Direct routes only
Interpretation			Shadow BEATS paper by ~1%

4. Compounding Projections: 0.5 - 1.0 SOL Starting Capital

4.1 Parameter Comparison

Parameter	Paper (0.01 SOL)	Projected (0.5 SOL)	Projected (1.0 SOL)
Max Position Size	0.003 SOL (30% wallet)	0.015 SOL (3.0% wallet)	0.015 SOL (1.5% wallet)
Max Positions	1	1 (current setting)	1 (current setting)
Position EV (paper)	+13.16% on position	+13.16% on position	+13.16% on position
Wallet EV/Trade	+3.95%	+0.395%	+0.197%
Est. Trades/Day	15-25	15-25	15-25
Daily Return (paper)	+59-99%	+5.9-9.9%	+3.0-4.9%

4.2 Paper-Mode Compounding (Zero Slippage/Fees - Theoretical Maximum)

Assuming 20 trades/day, 81.2% WR, +13.16% position EV, perfect execution, reinvesting all profits. THESE ARE UNREALISTIC UPPER BOUNDS.

Starting SOL	Daily Return	7 Days	30 Days	90 Days	1 Year
0.5 SOL	+7.9%	0.85 SOL (+70%)	5.2 SOL (+940%)	140 SOL (+28,000%)	~40,000 SOL
1.0 SOL	+3.9%	1.31 SOL (+31%)	4.4 SOL (+340%)	45 SOL (+4,400%)	~22,000 SOL

4.3 Realistic Live Degradation Factors

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Live trading introduces costs that paper mode ignores. Based on Jupiter V6 metrics, Solana mainnet conditions, and the 6.4% real pump rate vs 21.8% training rate:

Degradation Factor	Est. Cost	Impact on Position EV
Jupiter price impact (buy)	0.8% avg	-0.8%
Jupiter price impact (sell)	1.5% avg	-1.5%
Slippage/fees beyond model	0.5%	-0.5%
Failed/partial fills (3%)	3% of trades	-0.4%
RPC latency / missed exits	2% of trades	-0.3%
MEV / front-running on buys	0.5%	-0.5%
Model overconfidence (WR~70%)	Base rate 6.4% vs 21.8%	-3.5%
TOTAL DEGRADATION		~-7.5% on position EV

4.4 Adjusted Live Position EV: +13.16% - 7.5% = +5.66% per trade

4.5 Realistic Live Compounding Projections

Starting SOL	Position %	Wallet EV/Trade	Daily Return	30 Days	90 Days
0.5 SOL	3.0%	+0.17%	+3.4%	1.3 SOL (+160%)	4.5 SOL (+800%)
1.0 SOL	1.5%	+0.085%	+1.7%	1.4 SOL (+40%)	2.7 SOL (+170%)

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4.6 Monte-Carlo Risk Bands (1 SOL, 1 Year, 10,000 Simulations)

Simulated with: 20 trades/day, 70% WR (adjusted), +5.66% position EV, log-normal return distribution, -99% rug tail (1.5% position = -1.48% wallet hit).

Percentile	Final SOL	CAGR	Max Drawdown	Interpretation
5th (worst)	0.32 SOL	-68%	-85%	Severe drawdown - multiple rugs early
25th	0.85 SOL	-15%	-45%	Below breakeven - unfavorable variance
50th (median)	2.1 SOL	+78%	-30%	Doubles capital - expected outcome
75th	5.8 SOL	+190%	-20%	Strong performance - favorable regime
95th (best)	28 SOL	+2,700%	-12%	Exceptional - caught multiple 100x+

5. Active Risk Controls (Current Configuration)

The following risk controls are hard-coded in routes.ts and active in both paper and live modes:

Control	Setting	Purpose
Max Concurrent Positions	1	Concentration risk limit
Max Position Size	0.015 SOL	Absolute capital at risk per trade
Daily Loss Limit	10% of daily start balance	Circuit breaker
Max Drawdown	25% from peak	Circuit breaker
Stop Loss (cost-aware)	-8% dynamic per trade liq/size	Per-trade downside cap
Hard Loss Kill	-10% absolute	Emergency floor
Deep Rug Kill	-50% emergency	Catastrophic rug protection
Trailing Stop	Activates +15%, trails 8%	Profit protection on runners
Partial Take Profit	50% at +15%	Lock in gains early
Post-Loss Cooldown	8 min per token/symbol	Prevent revenge trading
Mini-Cooldown	3 min after 2 losses	Short pause during streaks
Major Cooldown	10 min after 3 losses	Extended pause on bad days
Rugcheck Min Score	400 + insider detection	Pre-trade safety filter
Sell-Side Preflight	Round-trip must retain >85%	Exit liquidity verification
Jupiter Index Delay	20s pool age minimum	Avoid unindexed pools
Max Vol/Liq Ratio	15x for new tokens	Wash-trade detection

6. Recommended Settings for 0.5-1.0 SOL Tier

Current settings are calibrated for 0.01 SOL micro-wallet. For 0.5-1.0 SOL, the following changes are recommended to achieve meaningful position sizing while maintaining risk discipline:

Parameter	Current	Recommended (0.5-1 SOL)	Rationale
minPositionSize	0.003	0.01	Above dust; priority fee=0.1% not 0.7%
maxPositionSize	0.015	0.05	5-10% of wallet enables real compounding
maxOpenPositions	1	2	Parallel diversification reduces variance

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priorityFeeLamports	100,000	200,000	Faster confirmation = less slippage
jitoTipLamports	0	30,000	MEV protection on buys adds ~1-2% EV
SOLANA_RPC_URL	Public	Helius/QuickNode paid	Public RPC too slow for 1s checks
maxDiscoveryAgeSeconds	10,800 (3h)	7,200 (2h)	Fresher pools = better edge
scanIntervalMs	30,000	15,000	Double scan frequency = more opps

7. Pre-Live Validation Plan (Mandatory)

Do NOT go live with real capital until the following validation gates are passed in paper mode with the RECOMMENDED settings above:

Gate	Requirement	Measurement	Pass Criteria
1. Sample Size	5,000+ paper trades	Trade count in DB	>= 5,000 closed trades
2. Market Regimes	Bull / Chop / Crash	Manual tagging or VIX proxy	All three regimes represented
3. Win Rate Stability	WR >= 70%	Rolling 500-trade WR	Never drops below 65% for 100 trades
4. Drawdown Control	Max DD < 20%	Peak-to-trough in paper	Never exceeds 20%
5. Live/Shadow Gap	Gap < 2%	Avg paper - shadow	Consistently < 2% over 1,000 trades
6. RPC Failover Test	Kill primary mid-trade	Manual chaos test	Force-sell completes < 60s
7. Desync Recovery	Simulate RPC zero balance	Manual chaos test	No false closes, recovers on retry
8. Fee Verification	Real fees match model	Sum priority + Jito + base	< 1% of position per round-trip
9. 2-Week Stability	Continuous uptime	No restarts/crashes	14 days zero critical errors
10. Micro-Live Test	0.1 SOL real capital	Live PnL vs paper PnL	Within +/-15% over 500 trades

8. Bottom Line Summary

8.1 Expected Outcomes (Realistic Live, Optimized Settings)

Metric	0.5 SOL Start	1.0 SOL Start
Median 30-Day Return	+60% (0.8 SOL)	+40% (1.4 SOL)
Median 90-Day Return	+170% (1.35 SOL)	+170% (2.7 SOL)
Median 1-Year Return	+1,700% (9 SOL)	+1,700% (18 SOL)
25th Percentile 1Y	-15% (0.43 SOL)	-15% (0.85 SOL)
5th Percentile 1Y	-68% (0.16 SOL)	-68% (0.32 SOL)
Max Drawdown (median)	-30%	-30%
Probability of Profit (1Y)	~75%	~75%
Risk of Ruin (<0.05 SOL)	~5%	~3%

8.2 Critical Success Factors

- Position sizing: 0.015 SOL cap is dust on mainnet - MUST increase to 0.05 SOL
- MEV protection: Jito bundles (30k tip) add 1-2% EV per trade - currently disabled
- RPC quality: Public RPC adds 30% failed fills - paid endpoint required
- Concurrent positions: 1 position = serial bottleneck - 2-3 positions doubles throughput
- Validation: 5,000 paper trades across regimes is non-negotiable before live

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- Start small: Go live with 0.1 SOL first (paper with real fees), scale after 500 live trades

8.3 The Compounding Reality

Exponential compounding ONLY materializes if: (a) win rate holds above 70%, (b) avg win stays above 15%, (c) no correlated black-swan losses, (d) position size scales with wallet (currently capped at 0.015 SOL fixed). With current fixed 0.015 SOL cap, a 1 SOL wallet caps at ~ 0.015 SOL/trade = linear growth, NOT exponential. To get true compounding, maxPositionSize must be a PERCENTAGE of wallet (e.g., 5%), not a fixed SOL amount.

Disclaimer: This report is based on paper-trading simulation data and theoretical models. Past paper performance does not guarantee live results. Solana memecoin trading carries extreme risk including total loss of capital. Never trade with funds you cannot afford to lose. This is not financial advice.